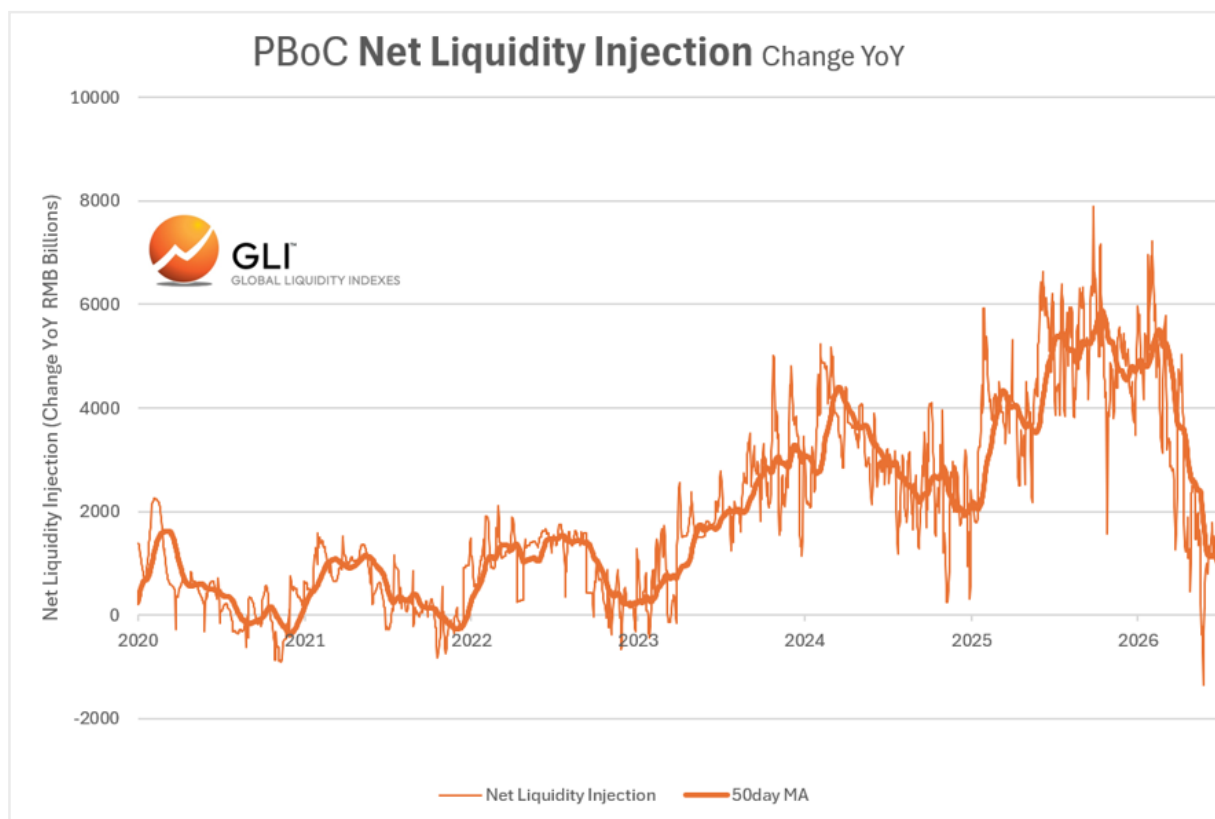




I need you to help analyze and reverse engineer a metric designed by Michael Howell for his china liquidity metric. He aggregates daily data and can see how much the PBOC is injecting into the money markets. This metric will help us analyze the commodity direction and buying of gold. What data is he using to create this chart and how can we create it on tradingview

Thought process

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I'll research how Howell constructs this before giving you a build recipe. Let me dig in.

Searched the web, loaded tools

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Here's the reverse-engineering, the data inputs, and three ways to rebuild it on TradingView — plus what your current chart is actually saying for copper/gold.

What Howell's metric actually measures

It's the **net liquidity the PBoC adds to (or drains from) the interbank system**, aggregated daily, then shown as a **year-over-year change** (RMB bn) with a **50-day moving average** to smooth the noise. "Net" and "YoY" are the two things that make it his, not a raw feed. The concept is effectively the high-frequency version of the **"Claims on Other Depository Corporations"** line on the PBoC balance

sheet — i.e., how much central-bank funding is outstanding to banks.

The key operation on every component is: **Net = new operations – maturing operations**. A CNY 900bn MLF with CNY 200bn maturing is a +700bn net injection, not +900bn. That netting is the whole game.

The data inputs he aggregates

He rolls up the daily/periodic PBoC operations, each netted against maturities:

- **7-day (and 14-day) reverse repos** — the daily OMO workhorse
- **Outright reverse repos** — newer tool, introduced Oct 2024
- **MLF** (Medium-term Lending Facility) — net of maturities
- **PSL** (Pledged Supplementary Lending) and **SLF**
- **Treasury bond purchases/sales** — PBoC began outright govvy operations Aug 2024
- **RRR changes** — reserve-requirement cuts release locked reserves (a liquidity add)
- **Government/fiscal deposit flows** at the PBoC (tax season drains, spending adds)

Sum the net of all of these = daily net liquidity provision. Take the 1-year change = his "Change YoY." Smooth with the 50-day MA.

Data source for the raw numbers: **PBoC's daily open-market-operation announcements** (pbc.gov.cn), with the maturity ladder tracked separately.

Practitioners pull this via **Wind/CEIC/Bloomberg** because you need the *maturity schedule* to net it, which the free feeds don't hand you cleanly.

Why TradingView can only approximate it

The relevant TradingView (TradingEconomics-sourced) symbols are:

- **ECONOMICS:CNLIVRR** — Liquidity Injections via Reverse Repo
- **ECONOMICS:CNLIVMLF** — Liquidity Injections via MLF
- **ECONOMICS:CNCBBS** — China Central Bank Balance Sheet
- plus Outright Reverse Repo, RRR, and Total Social Financing series

The catch: **these report the gross daily operation size, not net of maturities**.

(You can see it in the data — CNLIVRR prints tiny numbers like 0.5–5bn on quiet days.) TradingView's feed doesn't carry the maturity schedule, so you **cannot** reproduce Howell's true daily net series natively. You can only proxy it. Three options, in order of fidelity:

Option A — Balance-sheet YoY (closest to his concept, one symbol). Plot **ECONOMICS:CNCBBS** as a YoY change. Because the balance sheet is inherently *net*, this captures the real signal Howell is after. Downside: monthly, not daily.

Option B — Stacked-injection proxy (daily, directional). Combine the gross injection symbols and take a rolling net + YoY. This is the tedtalksmacro-style approach, extended to mimic his YoY view:

pirescript

//@version=6

indicator("PBoC Net Liquidity Proxy (YoY, 50D MA)", precision=0)

```
// gross daily injections (RMB bn) — these are GROSS, not net of maturities
rr = requestxsecurity("ECONOMICS:CNLIVRR", "D", close,
ignore_invalid_symbol=true)
mlf = requestxsecurity("ECONOMICS:CNLIVMLF", "D", close,
ignore_invalid_symbol=true)
```

```
grossDaily = nz(rr) + nz(mlf)
```

```
// approximate "net outstanding" as a trailing 1y cumulative of gross flows
netStockProxy = mathxsum(grossDaily, 252)
```

```
// Howell-style: YoY CHANGE of the stock, then 50-day smoothing
yoy = netStockProxy - netStockProxy[252]
yoyMA = ta.sma(yoy, 50)
```

```
plot(yoy, "Net Liquidity YoY (proxy)", color=colorxnew(colorxorange, 60),
style=plot.style_columns)
plot(yoyMA, "50D MA", color=colorxorange, linewidth=3)
hline(0)
```

Add **ECONOMICS:CNORR** (outright reverse repo) as a third input if it resolves for your account. Treat this as **shape/direction only** — the levels won't match his because of the gross-vs-net issue.

Option C — Build the real thing off-platform. Pull the daily OMO + maturity table (Wind/CEIC or scrape the PBoC announcements), compute net per day in a sheet, sum, take YoY, then import as a custom symbol via TradingView's Excel/CSV data-import (or just chart it in Excel). This is the only path to a faithful replica.

My recommendation: use **Option A as your "truth" signal** and **Option B as your daily early-warning overlay**.

Why this matters for copper and gold — and what your chart says now

Rising PBoC net liquidity (YoY) is a tailwind for industrial metals (China is the marginal copper buyer) and, via global liquidity, for gold. Your uploaded chart is the important part: the **50-day MA has collapsed from ~+5,000bn in 2025 toward ~+1,000bn, with the raw series briefly going negative (~-1,500bn) in early 2026**. That's a sharp **deceleration in the Chinese liquidity impulse** — a fresh headwind that lines up with everything from the last message: late-Boom regime, strong DXY, crowded copper longs. It **reinforces the cautious/sell-rips copper lean**. For gold it's more nuanced — gold can hold on the safe-haven/real-rate side even as the China impulse fades, so I'd treat a China-liquidity downturn as clearly copper-negative but only mildly gold-negative.

The bullish re-entry signal for copper would be this 50-day MA **turning back up through zero** — that's when the China impulse is re-accelerating.

Want me to save the Pine script as a file, and add copper + this proxy to a single

tracker so you get flagged when the China liquidity impulse inflects?

Sources: [tedtalksmacro PBoC Liquidity Injections \(TradingView\)](#), [China Liquidity Injections via Reverse Repo \(TradingEconomics\)](#), [China Liquidity Injections via MLF \(TradingEconomics\)](#), [China Central Bank Balance Sheet \(TradingView\)](#), [PBoC monetary policy tools — gov.cn](#), [RBA: China's Monetary Policy Framework](#)